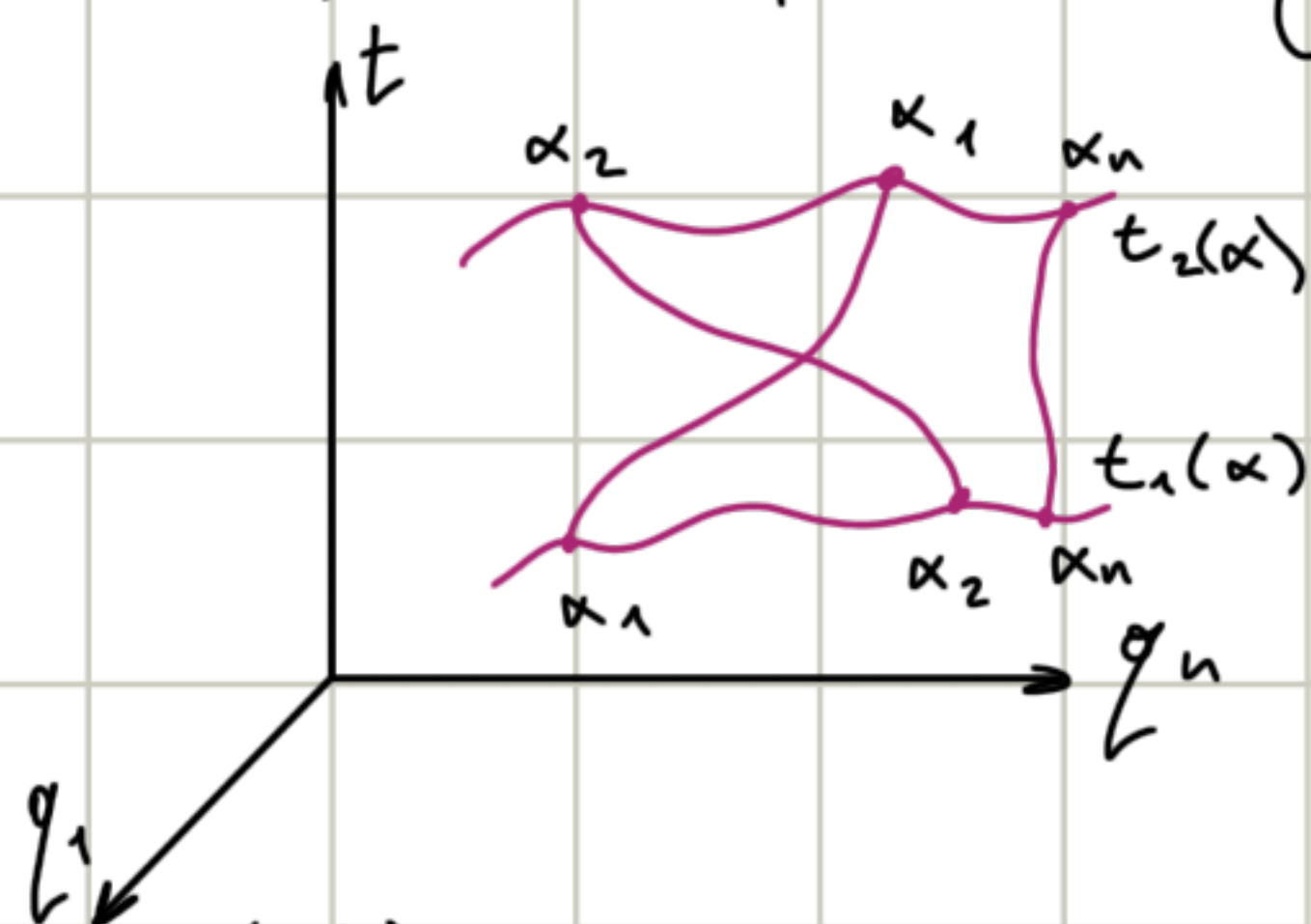


Универсальные уравнения

Общая ф-ла вариации q -виз по α .



$$W(\alpha) = \int_{t_1(\alpha)}^{t_2(\alpha)} L(q, \dot{q}, t) dt$$

$$\delta W(\alpha) = \frac{\partial W}{\partial \alpha} \delta \alpha$$

$$\delta W = L_2 \delta t_2 - L_1 \delta t_1 + \int_{t_1(\alpha)}^{t_2(\alpha)} \delta L dt$$

$$\delta t_2 = \frac{\partial t}{\partial \alpha} \delta \alpha \Big|_{t_2}$$

$$\int_{t_1(\alpha)}^{t_2(\alpha)} \delta L dt = \int_{t_1(\alpha)}^{t_2(\alpha)} \left(\frac{\partial L}{\partial q} \delta q + \frac{\partial L}{\partial \dot{q}} \delta \dot{q} \right) dt = \int_{t_1(\alpha)}^{t_2(\alpha)} \frac{\partial L}{\partial q} \delta q dt + \int_{t_1(\alpha)}^{t_2(\alpha)} \frac{\partial L}{\partial \dot{q}} \frac{d}{dt} (\delta q) dt$$

$$\delta \dot{q} = \frac{d}{dt} \delta q$$

$$\int_{t_1(\alpha)}^{t_2(\alpha)} \frac{\partial L}{\partial \dot{q}} d\delta q = \frac{\partial L}{\partial \dot{q}} \delta q \Big|_{t_1(\alpha)}^{t_2(\alpha)} - \int_{t_1(\alpha)}^{t_2(\alpha)} \delta q \frac{d}{dt} \frac{\partial L}{\partial \dot{q}} dt = \frac{\partial L}{\partial \dot{q}} \delta q \Big|_{t_1(\alpha)}^{t_2(\alpha)} - \int_{t_1(\alpha)}^{t_2(\alpha)} \left(\frac{d}{dt} \frac{\partial L}{\partial \dot{q}} \right) \delta q dt$$

$$\delta W = L_2 \delta t_2 - L_1 \delta t_1 + \int_{t_1(\alpha)}^{t_2(\alpha)} \frac{\partial L}{\partial q} \delta q dt - \int_{t_1(\alpha)}^{t_2(\alpha)} \frac{d}{dt} \frac{\partial L}{\partial \dot{q}} \delta q dt + \frac{\partial L}{\partial \dot{q}} \delta q \Big|_{t_1(\alpha)}^{t_2(\alpha)} \quad (*) \text{ обну.$$

$$\delta q \Big|_{t_1(\alpha)} = \frac{\partial q(t, \alpha)}{\partial \alpha} \Big|_{t=t_1(\alpha)} \delta \alpha$$

$$\delta q' = \frac{\partial q(t_1(\alpha), \alpha)}{\partial \alpha} \delta \alpha = \frac{\partial q(t_1(\alpha), \alpha)}{\partial t_1} \frac{\partial t_1}{\partial \alpha} \delta \alpha + \frac{\partial q(t_1(\alpha), \alpha)}{\partial \alpha} \delta \alpha = \dot{q}' \delta t_1 + \delta q \Big|_{t_1}$$

$$\delta W = L_2 \delta t_2 - L_1 \delta t_1 + \frac{\partial L}{\partial \dot{q}} \Big|_{t_2} \cdot (\delta q^2 - \dot{q}^2 \delta t_2) - \frac{\partial L}{\partial \dot{q}} \Big|_{t_1} \cdot (\delta q^1 - \dot{q}' \delta t_1) + \int_{t_1(\alpha)}^{t_2(\alpha)} \left(- \frac{d}{dt} \frac{\partial L}{\partial \dot{q}} + \frac{\partial L}{\partial q} \right) \delta q dt$$

$$\frac{\partial L}{\partial \dot{q}} \dot{q} - L = H$$

$$\delta W = -H_2 \delta t_2 + H_1 \delta t_1 + p^2 \delta q^2 - p^1 \delta q^1 + \int_{t_1(\alpha)}^{t_2(\alpha)} \left(\frac{\partial L}{\partial q} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}} \right) \delta q dt \quad (*) \text{ обну.}$$

$\int q(t, \alpha)$ - прямой путь

$$\delta W = p^i \delta q^i - H^i \delta t^i \Big|_1$$

Расширенное фазовое пространство



$$C_1: t_1 = t_1(\alpha), q_1 = q_1(\alpha), p_1 = p_1(\alpha)$$

$$H \Rightarrow q(t), p(t)$$

трубка прямых путей

$$t_1(0)=t_1(l), \quad q_1(0)=q_1(l), \quad p_1(0)=p_1(l)$$

$$c_2: t_2=t_2(\alpha), \quad q_2=q_2(\alpha), \quad p_2=p_2(\alpha)$$

$$\delta W = [p^2 \delta q^2 - H^2 \delta t_2] - [p^1 \delta q^1 - H^1 \delta t_1]$$

вдоль прямого пути

$$\int_{\alpha=0}^{\alpha=l} \delta W = W(l) - W(0) = 0$$

$$\int_{\alpha=0}^{\alpha=l} \delta W = \left(\int_{\alpha=0}^{\alpha=l} p^2 \delta q^2 - H_2 \delta t_2 \right) - \left(\int_{\alpha=0}^{\alpha=l} p^1 \delta q^1 - H_1 \delta t_1 \right) = \oint_{c_2} (p \delta q - H \delta t) - \oint_{c_1} (p \delta q - H \delta t) = 0$$

$$\oint_c (p \delta q - H \delta t) - \text{inv} \quad !$$

интеграл Пуанкаре - Кармана

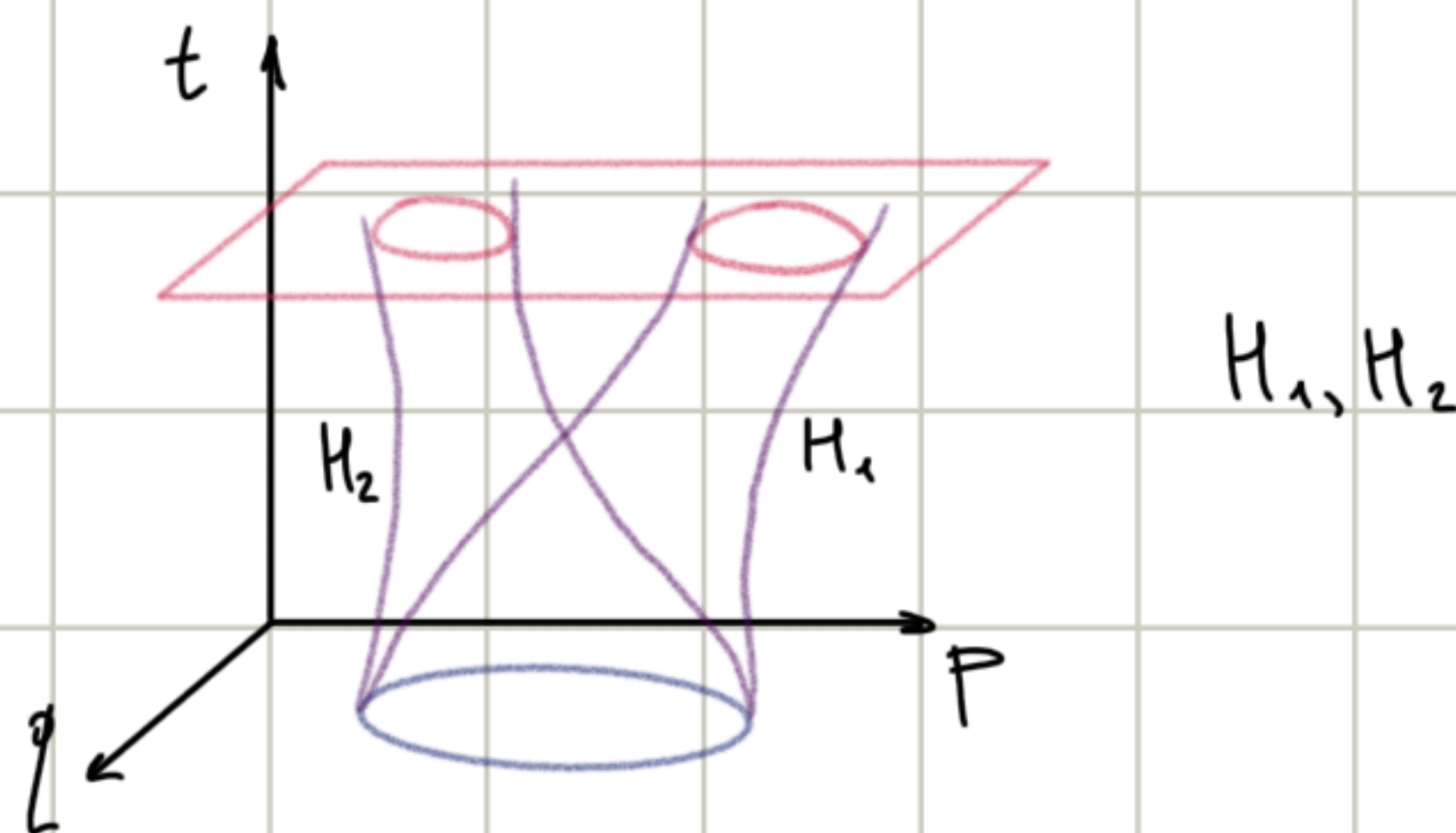
Универсальный инт. инв.

τ - изохрон. контур

$$\tau: t = \text{const} \quad \delta t = 0$$

$$\oint_{\tilde{c}} p \delta q = \text{inv}$$

инт. инв. Пуанкаре



Обратные th об и.и.

th

$$\left\{ \begin{array}{l} \dot{q}_i = Q_i(\bar{q}, \bar{p}, t) \\ \dot{p}_i = P_i(\bar{q}, \bar{p}, t) \end{array} \right. \quad (*)$$

и $\exists$ интер. выраж. $\oint_{\tilde{c}} p \delta q$ независ. от выбора изохр. $\tilde{c}$ вдоль трубки
прям. пути для (*), тогда (*) - гамильт., т.е. $\exists H^*$:

$$Q_i = \frac{\partial H^*}{\partial p_i}, \quad P_i = -\frac{\partial H^*}{\partial q_i}$$

и $\exists$ кроме этого $\exists F: \forall c \quad \oint_c (p \delta q - F \delta t) - \text{inv}$ для трубки и.и. (*), то

$$H^* = F + \frac{d\psi(t)}{dt}$$

$$\Delta. 1) I_\pi = \oint_{\tilde{c}} p \delta q = \text{inv}$$

$$\delta q^i = \underbrace{\dot{q}_i \delta t_i}_{=0} + \frac{\partial q}{\partial \alpha} d\alpha \Big|_i$$

$$\frac{dI}{dt} = 0 = \oint_{\tilde{c}} \frac{dp}{dt} \delta q + \oint_{\tilde{c}} p \frac{d}{dt} \delta q = \oint_{\tilde{c}} \frac{dp}{dt} \delta q + \oint_{\tilde{c}} p \delta \dot{q} = \oint_{\tilde{c}} \frac{dp}{dt} \delta q + p \dot{q} \Big|_{\tilde{c}} - \oint_{\tilde{c}} \dot{q} \delta p = \oint_{\tilde{c}} p \delta q - \oint_{\tilde{c}} Q \delta p = 0$$

$$\oint_{\tilde{c}} (p \delta q - Q \delta p) = 0$$

$$\exists \Phi: \frac{\partial \Phi}{\partial q} = p, \quad \frac{\partial \Phi}{\partial p} = -Q \quad ! \Rightarrow \Phi = -H^* \quad !$$

$$\oint_{\tilde{c}} \delta \Phi = 0$$

$$\frac{\partial^2 \Phi}{\partial p \partial q} = \frac{\partial p}{\partial p}, \quad \frac{\partial^2 \Phi}{\partial q \partial p} = -\frac{\partial Q}{\partial q}$$

$$2) \exists F:$$

$$\oint_c (p \delta q - F \delta t) = \oint_{\tilde{c}} p \delta q$$

$$\oint_{\tilde{c}} p \delta q = \oint_c (p \delta q - H^* \delta t)$$

$$\oint_c (F - H^*) \delta t = 0 = \oint_c d\psi$$

$$d\psi = \frac{\partial \psi}{\partial q} \delta q + \frac{\partial \psi}{\partial p} \delta p + \frac{\partial \psi}{\partial t} \delta t = (F - H^*) \delta t$$

$$\Rightarrow \frac{\partial \psi}{\partial q} = 0, \quad \frac{\partial \psi}{\partial p} = 0 \Rightarrow \psi = \psi(t)$$

$$H^* = F - \frac{d\psi(t)}{dt} \quad !$$

$$\mathcal{I}_1 = \oint (\sum A_i(\bar{p}, \bar{q}, t) \delta q_i + B_i(\bar{p}, \bar{q}, t) \delta p_i)$$

th du XyauMyra

Если $\mathcal{I}_1$ -универс. и.и., то он $\mathcal{I}_1 = c I_\pi$

Δ. B-Bo гур 1d мур.

$$\exists \mathcal{I}_1 = \oint_{\tilde{c}} A \delta q + B \delta p \quad \text{-inv} \quad (\forall H)$$

$$\frac{d\mathcal{I}_1}{dt} = 0 = \oint_{\tilde{c}} \frac{dA}{dt} \delta q + A \dot{q} + \oint_{\tilde{c}} \frac{dB}{dt} \delta p + B \dot{p} = \oint_{\tilde{c}} \frac{dA}{dt} \delta q - \dot{q} \delta A + \frac{dB}{dt} \delta p - \dot{p} \delta B \quad \ominus$$

$$\oint_{\tilde{c}} A \dot{q} = A \dot{q} \Big|_{\tilde{c}} - \oint_{\tilde{c}} \dot{q} \delta A$$

$$\ominus \oint_{\tilde{c}} \left[\left(\frac{\partial A}{\partial q} \dot{q} + \frac{\partial A}{\partial p} \dot{p} + \frac{\partial A}{\partial t} \right) \delta q - \dot{q} \left(\frac{\partial A}{\partial q} \delta q + \frac{\partial A}{\partial p} \delta p \right) + \left(\frac{\partial B}{\partial q} \dot{q} + \frac{\partial B}{\partial p} \dot{p} + \frac{\partial B}{\partial t} \right) \delta p - \dot{p} \left(\frac{\partial B}{\partial q} \delta q + \frac{\partial B}{\partial p} \delta p \right) \right] =$$

$$= \oint_{\tilde{c}} \delta q \left[\frac{\partial A}{\partial p} \dot{p} + \frac{\partial A}{\partial t} - \dot{p} \frac{\partial B}{\partial q} \right] + \delta p \left[-\frac{\partial A}{\partial p} \dot{q} + \frac{\partial B}{\partial q} \dot{q} + \frac{\partial B}{\partial t} \right] = 0$$

$$\begin{cases} \dot{q} = \frac{\partial H}{\partial p} \\ \dot{p} = -\frac{\partial H}{\partial q} \end{cases}$$

$$\exists \Phi: \quad \frac{\partial \Phi}{\partial q} = \left[\frac{\partial A}{\partial p} - \frac{\partial B}{\partial q} \right] \cdot \left(-\frac{\partial H}{\partial q} \right) + \frac{\partial A}{\partial t}$$

$$\frac{\partial \Phi}{\partial p} = \left[\frac{\partial A}{\partial p} - \frac{\partial B}{\partial q} \right] \cdot \left(-\frac{\partial H}{\partial p} \right) + \frac{\partial B}{\partial t}$$

$$-\frac{\partial u}{\partial p} \frac{\partial H}{\partial q} - u \frac{\partial^2 H}{\partial p \partial q} + \frac{\partial^2 A}{\partial p \partial t} = -\frac{\partial u}{\partial q} \frac{\partial H}{\partial p} - u \frac{\partial^2 H}{\partial q \partial p} + \frac{\partial^2 B}{\partial q \partial t}$$

$$-\frac{\partial u}{\partial p} \frac{\partial H}{\partial q} + \frac{\partial u}{\partial q} \frac{\partial H}{\partial p} + \frac{\partial}{\partial t}(u) = 0$$

$$\exists H=0 \Rightarrow \frac{\partial u}{\partial t} = 0 \Rightarrow u = u(p, q)$$

$$\exists H=q \Rightarrow -\frac{\partial u}{\partial p} = 0 \Rightarrow u = u(q)$$

$$\exists H=p \Rightarrow \frac{\partial u}{\partial q} = 0 \Rightarrow u = c!$$

$$\frac{\partial A}{\partial p} - \frac{\partial B}{\partial q} = c \Rightarrow \frac{\partial (A - cp)}{\partial p} - \frac{\partial B}{\partial q} = 0$$

$$\oint_{\tilde{c}} (A - cp) \delta q + B \delta p = 0$$

$$\exists \psi: \quad \frac{\partial \psi}{\partial q} = A - cp \quad \frac{\partial \psi}{\partial p} = B$$

$$\frac{\partial (A - cp)}{\partial p} = \frac{\partial B}{\partial q}$$

$$\oint_{\tilde{c}} A \delta q + B \delta p = c \oint_{\tilde{c}} p \delta q \quad \blacktriangle$$